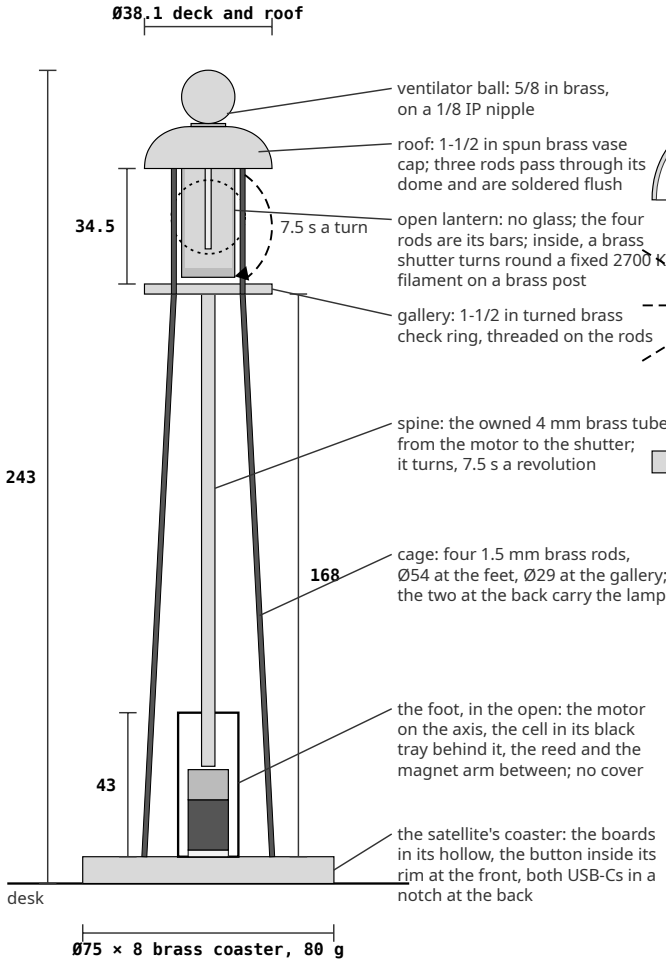
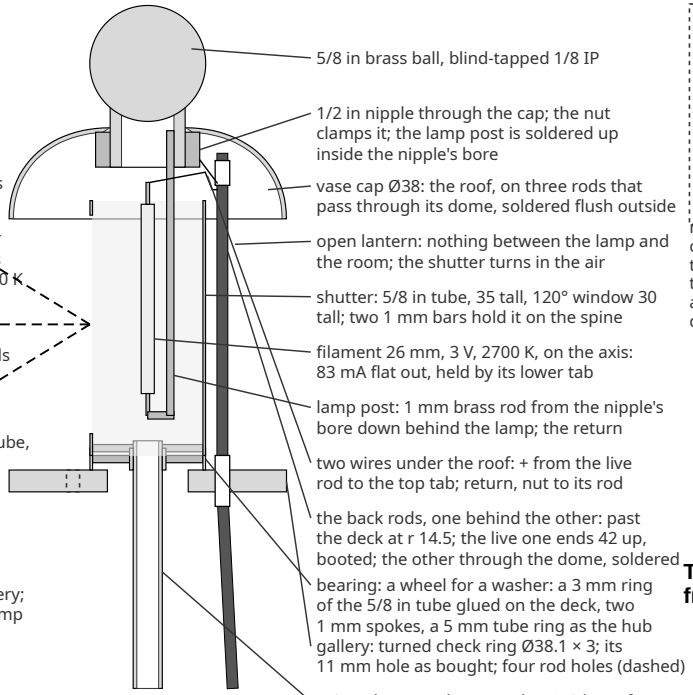
Desk lighthouse: how the finished thing looks (Rev D: the open wire frame, on a cell)

Four brass rods from a brass coaster to a brass roof, a turning spine up the middle, the motor and the cell standing in the open at the foot, an open lantern on top: no glass, no house. 243 mm tall, Ø75 at the base, about 217 g before the ba

Front view



Lantern section, front on the left; the two front rods are cut away



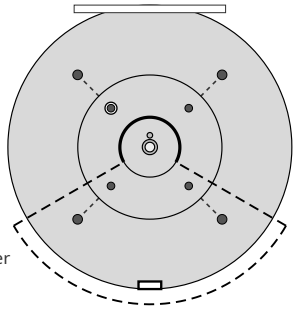
How it behaves

off	lamp dark, shutter closed (brass to you), boost asleep
lamp	shutter parked open to you, filament 70 % (58 mA)
beacon	the shutter turns, 7.5 s a revolution: a 2.5 s sweep, then brass
button	in the rim at the front. Short press: rotation on/off (off parks the window on you); long: lamp off, low, full (off closes the shutter)
cell	about 7 h of beacon, 8 of lamp; low: dim, park, sleep
charge	USB-C at the back, 2.3 h; it runs from the cable meanwhile

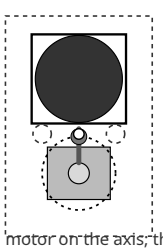
Key numbers

243 tall · Ø75 base · cage Ø54 → Ø29 over 168 · lantern open 34.5
217 g, centre of mass 81 up, tips at 25° (0.3 N at the ball);
with 80 g of steel in the coaster: 60 up, 32° (0.5 N)
from the cell: 129 mA beacon (7 h), 112 lamp (8 h), 20 asleep
rail 70 mA typical, 119 peak · lamp 83 mA (249 mW) · charge 500 mA
cuts: rods 4 × 223 · spine 148 · shutter 35 · post 42 · 1 mm rod 64 · two 3 mm rings

Plan, roof off



The foot, looking down

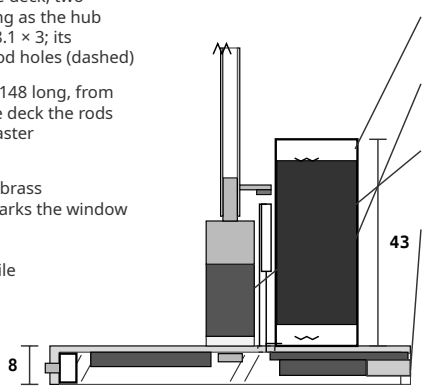


motor on the axis; the arm on the spine, its magnet over the reed behind the motor; the cell in its black tray at the back; two wire holes dashed; no house: all open

front at the bottom; the 120° beam to you
coaster Ø75; the four feet at r 27 (Ø54), leaning in
deck Ø38.1; the rods pass it at r 14.5 (Ø29)
shutter Ø15.9, open 120° to the front, on two bars
spine Ø4 in the wheel's hub · post at r 3.1, behind

no house: motor, tray, reed and arm on the bare brass
the live rod, back-left, ringed white: its sleeve
notch 40 wide in the rim at the back: charge and flash
the button in the rim at the front: the black mark
turns 8 rpm: 7.5 s a revolution

The foot and the coaster, cut front on the left



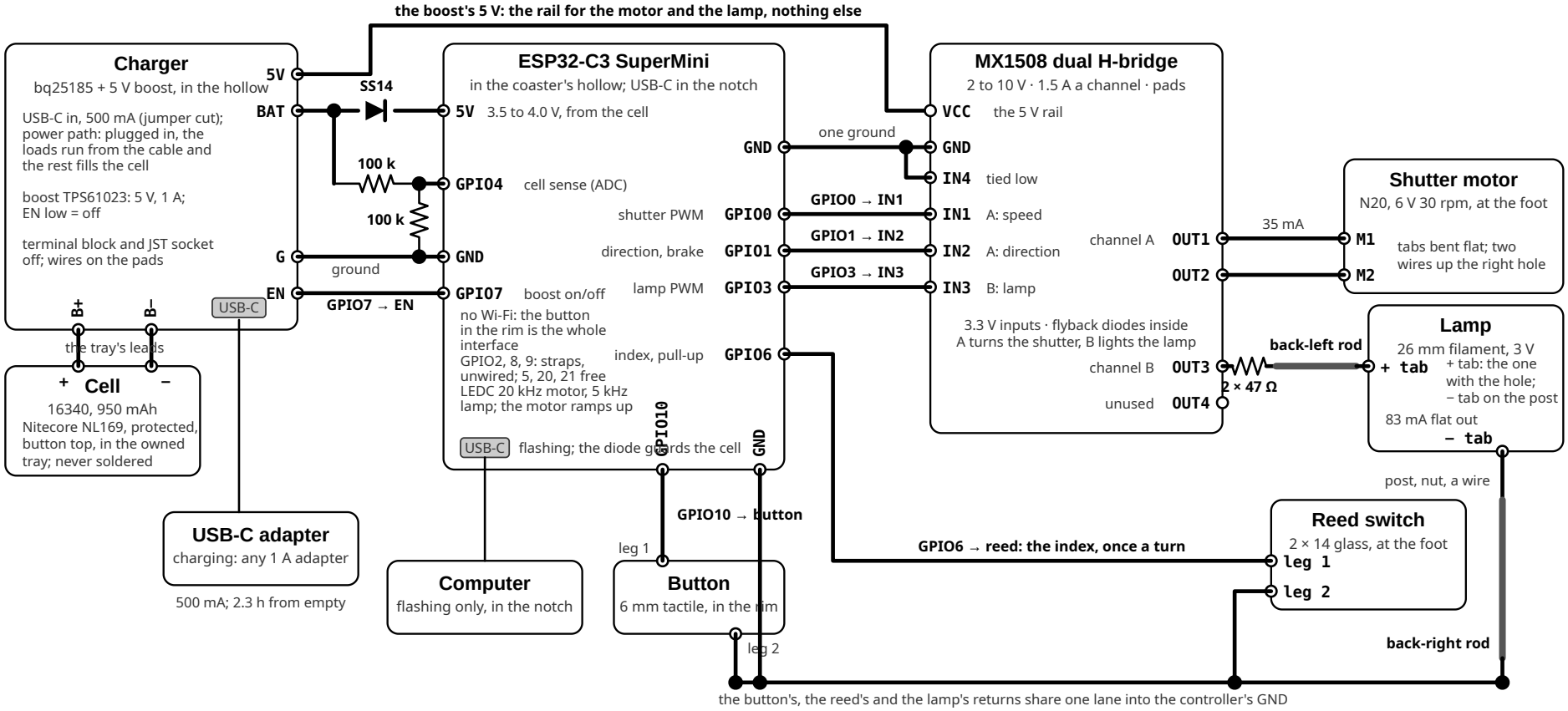
no house: the motor, the tray, the reed and the magnet arm stand in the open on the coaster's top
cell: a 16340 standing in the owned black tray behind the motor; its leads down the left hole
N20 motor on a 2 mm bed, shaft or the axis; the spine over its D shaft; the magnet on a 1 mm arm over the reed behind the motor
charger in the hollow, glued up to the ceiling, USB-C in the notch; the controller beside it, cut away; driver in front; washers fill the rest

left: the button inside the rim at the front, its plunger through a Ø3.6 hole 3.4 mm up from the bottom edge; press the front of the coaster

Rev D, drawn 2026-11-09 evening for approval, not built. The check ring's thickness, the vase cap's height and the coaster's hollow are assumed: measure them and rerun the script. Sources: the listings and the pages named in the record.

Desk lighthouse: every connection on one sheet, the power chain first

A dot is a joint. A line that hops over another does not touch it. Thick = 30 AWG wire you solder. Thin = a lead or a cable. Brass-filled = a rod of the frame, carrying current. Left to right: cell, charger, controller, driver, loads. Zero crossings.



The budget: from the cell, and on the boost's 5 V rail

cell	129 mA with the beacon turning and the lamp at 70 %: about 7 h · 112 mA lamp alone: 8 h · 20 mA asleep, boost off: 2 days
rail	70 mA typical on the boost's 5 V, 119 at most in normal running; the boost stalls if more than 200 mA is switched on in a step, so the motor ramps
motor	35 turning at 32 % duty (8 rpm); about 400 if the shutter jams, which the reed's silence reports
lamp	83 flat out, 58 at the 70 % default; 249 mW in the lantern, 81 mW in each resistor; up one rod, down another
controller	20 from the cell through the diode, no Wi-Fi; the divider 21 µA
charge	500 mA with the jumper cut, about 2.3 h from empty; plugged in, the lighthouse runs from the cable and the cell fills with the rest

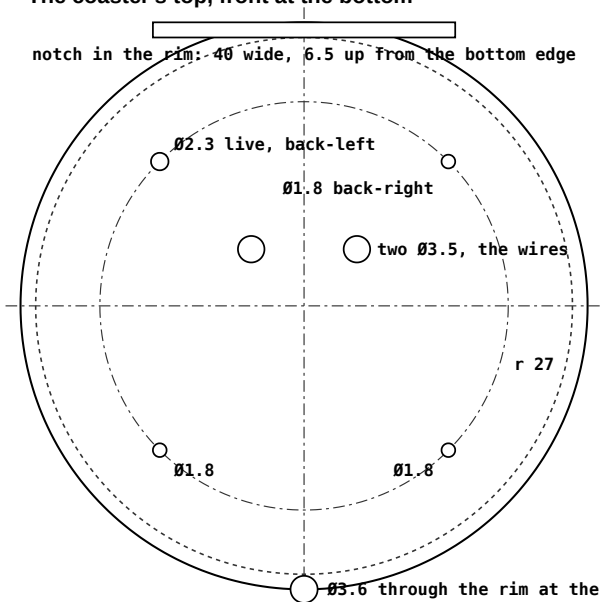
Never: the cell without the diode between it and the controller · the charger's jumper uncut · the lamp without its resistors · the live rod touching any other brass (meter it) · an iron near the cell or the magnet.

Rev D, drawn 2026-11-09 evening for approval, not built. 0 crossing(s) on this sheet. Sources: the charger's pinouts page, the SuperMini notes (5V = VBUS), the MX1508 datasheet, the satellite's record of the same cell, charger and diode, the

Desk lighthouse: templates at actual size

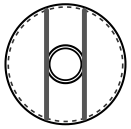
Print landscape on Letter at Actual size: the 50 mm bar measures 50 mm. Prick the hole centres through the paper onto the brass; drill with the cordless drill at its slowest, saw with the jeweler's saw, file to the line.

The coaster's top, front at the bottom

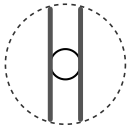


The feet sit on the Ø54 circle at 45°; the wire holes sit between the motor and the tray, 7 either side of the axis and 7.5 behind it. Drill the 1.8, 2.3 and 3.5 straight through the 1.3 mm top; the Ø3.6 for the button goes through the rim at the front. Saw the 40 mm notch from the rim's bottom edge at the back with two cuts, break it out, file.

The spine's hardware: 1 mm rod and tube rings, actual size



the bearing wheel, on the deck
rim: a 3 mm ring of the 5/8 in tube
hub: a 3 mm ring of the 5 mm tube
two spokes 14.3, tangent to the hub



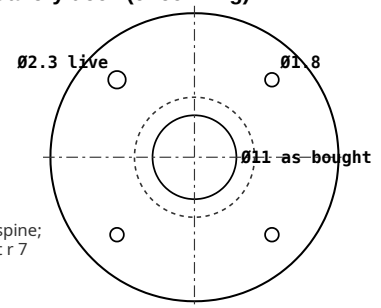
the shutter's floor
two bars 14.6, tangent to the spine,
0.5 mm up inside the drum (dashed)



the magnet arm
6.5 of 1 mm rod on the spine;
the magnet under it at r 7

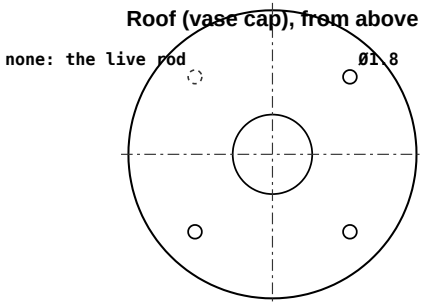
Five pieces of 1 mm rod (64 mm in all) and two rings; file the ring ends square.
Make the wheel on the spine itself: the hub ring on the spine, the spokes soldered to its flanks,
the rim ring to the spoke ends; it centres itself. Slide it down onto the deck and glue the rim.
The bars go on last, with the drum: solder them to the spine's flanks, then their ends to the wall.

Gallery deck (check ring)



The four rods pass at r 14.5, at 45° (front at the bottom).
Dashed: the wheel's rim Ø15.9 over the Ø11 hole;
the live rod's hole, back-left, takes its sleeve.

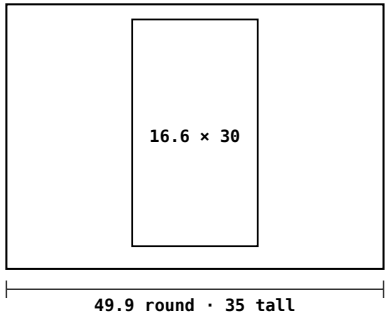
Roof (vase cap), from above



Three Ø1.8 holes at r 14.5, on the dome's slope:
centre-punch, drill from outside; the rods pass
through and are soldered flush. Measure the cap first.

50 mm: check it with a ruler

The shutter, unrolled: 5/8 in brass tube



120° window, 3 up from the bottom end. Wrap the strip round the tube, prick the corners, saw, file.
A 3 mm cut of the same tube is the wheel's rim.

One rod, not to scale



cut 223 · 6 mm foot · deck mark 174.5 from the bend
live rod: sleeves at foot and deck, boot at the tip, 1.4 shorter

Order of work

1. Coaster: four rod holes, two wire holes, the notch at the back, the button's Ø3.6 in the rim at the front. Roof: three Ø1.8 holes in the dome. Shutter: cut 35, saw the window, plus a 3 mm ring from the same tube. Charger: jumper cut.
2. From 1 mm rod: two bars, two spokes, the arm; a 3 mm ring of the 5 mm tube. Make the wheel on the spine as a jig: hub ring, spokes on its flanks, rim ring on their ends. Solder the arm to the spine 6.5 above where the gearbox face sits.
3. Bend the motor's tabs, solder its wires, set it on its bed, shaft on the axis. Slide the spine over the D shaft with a drop of gel glue; sight it true as the motor turns slowly. Glue the magnet under the arm and the reed behind the motor.
4. Rods: sleeve the live one; thread each through the coaster hole, bend the foot under, glue. Slide the deck down to 168, nudge the rods vertical and glue it; slide the wheel down onto the deck, glue its rim over the hole; trim the live rod.
5. Shutter: solder the bars to the spine's flanks 0.5 mm inside the drum's bottom edge, window to the front at the index, then the bar ends to the wall from below. Lantern: post into the nipple, filament on the post, nut, then the cap down over the three rods' tips, soldered flush outside, the live rod booted beneath; the ball; two wires to the back rods.
6. Boards, diode, divider, the button inside the rim, the washers in the hollow, both ports in the notch. The tray behind the motor on a dot of glue, leads down the left hole; the cell in. Flash through the notch; then the meter checks.

Rev D, drawn 2026-11-09 evening for approval. The deck assumes the 1-1/2 in check ring, the roof the 1-1/2 in vase cap, and the coaster's top 1.3 mm from its listed weight. Measure the real parts and rerun the script before cutting.

DESK LIGHTHOUSE · STEP 1: THE FINAL BUILD · REV D, THE OPEN WIRE FRAME ON A CELL, DRAWN 2026-11-09 EVENING FOR APPROVAL

A lighthouse drawn in four brass rods, with a lamp that stands still while the brass turns, on a cell, and nothing else

How it looks, how it is wired and powered, every part and every pin. 243 mm tall on a Ø75 brass coaster, about 217 g before the ballast, about 7 hours of beacon on a charge. Paper design of 2026-11-09, after Rev A of 2026-11-07, Rev B of 2026-11-08 and Rev C of 2026-11-09 (approved as drawn, then its glass and house rejected the same evening). The science and engineering behind it is Desk Lighthouse Physics. The Amazon parts are ordered; nothing built.

What it is

A lighthouse for the desk drawn in wire, 243 mm tall, on a cell, and nothing else. Four 1.5 mm brass rods rise from the same brass coaster the satellite stands on, Ø54 at their feet and Ø29 where they pass the gallery, then straight up as the lantern's bars to the roof, which three of them carry. Up the middle a 4 mm brass tube, the spine, turns slowly from a gearmotor standing at the foot. The lantern is open: a brass gallery (a lamp-maker's check ring) below, a brass roof (a vase cap) above with a brass ball for the ventilator, and air between. In it a warm-white LED filament stands still on the axis, and a brass shutter cut from a 5/8 in tube turns round it on the spine: one revolution in 7.5 s, a 2.5 s sweep of light across the room, then brass. A reed switch and a magnet on a thin brass arm at the foot tell the firmware where the window is, so when you turn rotation off it parks facing you, and when the lamp goes off it closes.

At the foot the motor stands on the axis, the family's Nitecore 16340 cell stands in the owned black holder behind it, and the reed stands between them, all in the open. The charger board, the owned ESP32-C3 SuperMini and the owned MX1508 driver lie in the coaster's hollow, the charger's USB-C and the controller's side by side in a notch at the back: plug a cable in there to charge, or to flash. The one button hides inside the rim at the front; press the coaster. About 7 hours of beacon on a charge, 2.3 to fill. The lamp's current goes up one rod and comes down another. No screen, no glass, no house, no paint.

The reference is Huy Vector's Lighthouse Desk Lamp: a parts list (brass sheet and tube, copper wire, a 3 V 30 rpm gear motor, a 39 mm Edison filament, a glass fuse, a Type-C charging module, a battery, a bamboo coaster) and no more. Ours keeps the idea, a small brass lighthouse on a coaster with a motor, a filament and a cell, and is its own build otherwise.

This page is step 1 of the house rule: the final build first. Rev C was approved by the builder on 2026-11-09 as drawn; that evening the builder rejected the acrylic glass, the brass sheet and any plastic house, and this is Rev D, drawn 2026-11-09 evening for approval: the same lighthouse with an open lantern, no house, and its three small brass pieces made from rod and tube. One form sheet, one wiring sheet with the power chain first, one sheet of templates, every part and every pin. Brass at the builder's word for this build. The science and engineering behind it (a fixed lamp and a turning window, a filament behind a resistor and why the rail is 5 V, a gearmotor at a third of its voltage, the budget from the cell, why the rods lean, a magnet and a reed) is step 2, Desk Lighthouse Physics. The Amazon parts are ordered; nothing is cut or soldered.

The decisions

A LIGHTHOUSE DRAWN IN FOUR LINES

Four 1.5 mm brass rods run from the coaster to the roof: Ø54 at the feet, leaning in 4.3° to Ø29 where they pass the gallery, then straight up as the lantern's bars to the roof, which three of them carry. No rings, no braces, no glass, no house: the coaster holds their feet, the deck holds their waists, the roof their tips, and a 1.5 mm brass rod fixed at both ends carries the lantern's 67 g with a buckling margin of about 211 to one. Up the middle the 4 mm spine turns.

243 mm tall, about 217 g before the ballast. Rev A's tube and cube, Rev B's screen and Rev C's glass and house are gone; what is left is a wire drawing of a lighthouse with a lamp in it.

THE LAMP STANDS STILL AND THE BRASS TURNS ROUND IT

A real lighthouse does not spin its lamp; the lens turns round a fixed light. Here the lamp is a 26 mm LED filament standing on the axis, and a brass drum with a 120° window turns round it, one revolution in 7.5 s: a 2.5 s sweep of light across the room, then brass. No slip rings, no wire ever moves. When rotation is off, the window parks facing you and the lantern is a lamp. When the lamp is off, the shutter closes and shows you brass.

ON A CELL: THE BATTERY STANDS AT THE FOOT

The builder asked for a battery and a charger below (2026-11-09). The battery is the family's cell, a Nitecore 16340 of which thirteen are in the drawer, in the owned holder, standing on end on the coaster's top behind the motor, in the open: 18 × 17 × 43 mm of black tray, the one dark block in the build, a colour the rule allows. The cell lifts out of its tray; nothing is soldered to it. 129 mA from the cell with the beacon turning and the lamp at 70 %: about 7 hours of lighthouse, 8 of lamp alone, 2 days asleep with the boost off. The cell's voltage comes to the controller through two 100 kΩ, so it dims, parks and shuts down in order instead of dying mid-turn.

THE CHARGER IN THE COASTER, THE PORTS AT THE BACK

The same charger board as Cyber Dawn and the satellite, a bq25185 with a 5 V boost: USB-C in, 500 mA into the cell with its jumper cut (2.3 h from empty), a power path that runs the lighthouse from the cable while it charges, and a 1 A boost that makes the 5 V rail for the motor and the lamp. The controller takes the cell straight through an SS14, the satellite's trick, so the boost can sleep: GPIO7 wakes it only while something moves. Its terminal block and JST socket come off and it lies 4.8 mm flat in the coaster's hollow beside the controller, both ports in one 40 mm notch at the back of the rim: the charging cable and, rarely, the flashing cable. That notch is the build's one compromise.

THE MOTOR ON THE COASTER, THE SPINE UP THE MIDDLE

A wire frame has nowhere to hide a motor at the top, so the motor stands at the foot on a bed of hot glue, shaft up on the axis, and the owned 4 mm brass tube slides over its D shaft and runs 148 mm up through the deck's bearing to the shutter. The bearing is a wheel made of what is already here: a 3 mm ring of the 5/8 in tube glued on the deck over its 11 mm hole, two 1 mm spokes, and a 3 mm ring of the owned 5 mm tube as the hub, which the 4 mm spine telescopes into with about 0.1 mm of play. It is soldered up on the spine itself, so it centres itself, then slid down onto the deck. The spine is the one line beyond the outline, and it is the one that moves: the lighthouse's own lamp post, turning.

A REED SWITCH SO IT KNOWS WHERE THE WINDOW IS

A DC gearmotor has no idea where it stopped. A 1 mm brass rod 6.5 mm long is soldered to the spine as an arm, 6.5 mm above the gearbox, with a 3 × 1 mm magnet glued under its tip, 7 mm out, placed opposite the shutter's window; a glass reed switch stands behind the motor, in front of the cell, with its top 2 mm under the magnet's path, so it closes once a revolution at the moment the window faces front. From that one pulse the firmware times each revolution and trims the PWM to 7.5 s, parks the window on you when you turn rotation off, closes the shutter by running on half a turn when the lamp goes off, and notices a jam. Two short wires; nothing climbs the tower for it.

TWO OF THE FOUR RODS ARE THE LAMP'S WIRES

Nothing may climb a wire frame but the frame, so the lamp's current goes up one rod and comes down another. The back-left rod is live: white heat-shrink where it passes the coaster and the deck, a white boot on its tip, a wire soldered to its foot in the hollow and another from its tip to the filament's + tab under the roof; it is the one rod that stops short of the roof. The back-right rod is the return, soldered to a ground wire at its foot and to the nut's wire at its tip, and through the roof like the two front rods. 8.1 mΩ each; the brass does not notice 83 mA. The rover's frame is its ground; this is the same idea with one rod insulated, and the meter checks it before power.

A LAMP POST THAT IS ALSO THE RETURN

A rigid LED filament has a metal tab at each end, the + one with a hole in it. So a 1 mm brass rod is soldered 5 mm up inside the nipple's hollow bore, runs down behind the lamp at r 3.1 inside the shutter, bends forward at the bottom and is soldered to the lower tab: it holds the filament rigid on the axis and carries its return to the nut. The + tab at the top takes one wire. Inside the drum a glowing strip and a thin brass post behind it, the way a lamp sits in a real lantern.

WARM WHITE, 2700 K, NOT THE FAMILY'S AMBER

The hologram and the satellite use a 2200 K filament as a pool of amber on brass, and eight of them are spare. A lighthouse lamp is not a pool: it is the light itself, and the rule is white or warm white. A 26 mm, 3 V, 100 mA filament at 2700 K is an incandescent lamp's colour; 83 mA through 23.5 Ω from the 5 V rail, 249 mW, dimmed to 70 % by day. Its phosphor is yellow when off, which is why the shutter closes over it.

NO HOUSE, NO GLASS, NO SCREEN

The screen went at the builder's word on the morning of 2026-11-09; the acrylic lantern glass and the brass-sheet house went that evening ("so ugly"; "just use the wire frame as a lighthouse, nothing else"). So the lantern is open: the four rods are its bars, the shutter turns in the air, and nothing optical is lost, since the glass was a window and not a lens. At the foot the motor, the reed, the magnet arm and the cell in its tray stand on the brass in the open: steel, glass, brass and black, every one a colour the rule allows. Nothing is folded, nothing is painted, nothing is glued over anything.

THE THREE SMALL BRASS PIECES ARE ROD AND TUBE, NOT SHEET

Rev C cut three discs from 0.4 mm sheet: the shutter's floor, the magnet's collar and the deck's bushing. Rev D makes each from the rod and tube already on the bench. Two 1 mm bars 14.6 mm long lie across the shutter's bottom, 0.5 mm up, soldered to the spine's flanks and the drum's wall: the floor. One 1 mm arm on the spine carries the magnet: the collar. A wheel of two tube rings and two spokes sits on the deck: the bushing. Lighter, nothing to buy, and all of it in the same wire language as the cage.

LAMP-MAKER'S PARTS AS ARCHITECTURE

The gallery is a 1-1/2 in turned brass check ring, a solid disc with an 11 mm hole already where the spine goes, threaded down the rods. The roof is a 1-1/2 in spun brass vase cap, held by the rods themselves: three of them pass through its dome at r 14.5 and are soldered flush from outside, the one joint that closes the cage at the top. The ventilator ball is a 5/8 in brass ball tapped 1/8 IP, on a 1/2 in nipple with a beaded nut inside the cap that also carries the lamp post. Five parts, about $7, from one lamp-parts house, each already the right shape and the right metal, none of it machined. Nothing is printed.

BRASS LINES AND WHITE LIGHT

The brief was a brass build with some ivory; the ivory went with the house, and what is left is brass lines and warm white light. The lines are brass: the rods, the spine, the deck, the shutter, the cap, the ball, the coaster. Black only in the pad under the coaster, the cell's wrap and its tray, which stand at the foot in the open as the colour rule allows; the motor is steel, the reed glass; the live rod's sleeves are white heat-shrink. The boards live in the coaster's hollow, so nothing coloured shows. Brass at the builder's word for this build; the family's rule since the drone is stainless.

ONE OWNED DRIVER BOARD DOES BOTH JOBS

The camera Pocket AI's spare MX1508 dual H-bridge: channel A is the shutter's motor with speed, brake and the parking move; channel B, with one input tied low, is a 1.5 A high-side switch for the lamp through its resistors, the lamp returning through the frame. Both loads get the boost's 5 V, the controller's GPIOs see only logic, and there are no MOSFETs, gate resistors or flyback diodes to wire. Its header and screw terminals come off so it lies 3 mm flat in the hollow.

THE COASTER IS THE SATELLITE'S, AND IT IS THE BOX

The same 75 × 8 mm brass coaster the satellite stands on, so the two match on the desk; the one bought on 2026-11-06 for the satellite became this build's on 2026-11-09 and arrived that day, and the satellite's comes later. Its listing's 80 g says a top about 1.3 mm thick over a hollow about 6.7 deep: enough for the charger (4.8 mm stripped of its terminal block), the controller (5.3 mm with its port) and the driver glued up to the ceiling, the button glued inside the rim at the front with its plunger through a 3.6 mm hole, the rods' feet bent under the top, and about 80 g of steel washers beside them. Alone it tips at 25°, a 0.3 N push at the ball; weighted, the centre of mass drops to 60 mm and the tip to 32°, 0.5 N; the cell low at the foot helps. Measure the hollow.

A REFERENCE, NOT A COPY

Huy Vector's page is a parts list: 0.2 mm brass sheet, 4 mm brass tube, copper wire, a 3 V 30 rpm gear motor, a 39 mm Edison filament, a glass fuse, a Type-C charging module, a battery, a bamboo coaster. Ours keeps the idea (a small brass lighthouse on a coaster, a gear motor turning something in the lantern, a filament lamp, a cell and a charging port, brass) and does the rest its own way: a wire tower instead of rolled sheet, a fixed lamp and a turning brass shutter, a reed index so it parks, the family's cell and charger, lamp parts for the lantern, no sheet at all. Side by side they share a silhouette, a motor and a battery.

HOW IT SITS WITH THE FAMILY

Brass rods and a few rings are the Pocket AI Sphere's language; an open wire frame that carries current is the rover's; the coaster, the cell, the holder, the charger and the diode are the satellite's, wired the same way. At 243 mm it is the tall one; the satellite is 159, Cyber Dawn 82, the hologram 72. Few lines: four rods and a spine, and no closed surface at all, the first build in the family with nothing to lift off. Prototype, not product: the cell lifts out of its tray, the firmware reflashes through the notch, and the roof comes off with a hot iron on three joints.

Dimensions

WHAT	MM
overall height, desk to the top of the ball	242.8
brass coaster, diameter	75.0
brass coaster, height	8.0
coaster's top plate, thickness (from its weight; measure)	1.3
coaster's hollow, depth (measure)	6.7
notch in the rim at the back, width (both USB-C ports)	40.0
notch, height from the bottom edge	6.5
the two wire holes in the top, between the motor and the tray	3.5
wire holes, either side of the axis	7.0
wire holes, behind the axis	7.5
the button's hole in the rim at the front	3.6
button hole, up from the bottom edge	3.4
cage at the feet, across	54.0
cage at the gallery, across	29.0
rod holes in the coaster, from the axis	27.0

rods at the deck and above, from the axis	14.5
coaster top to the deck's underside	168.0
rods' lean below the deck, degrees	4.3
each rod, cut length	223.0
deck mark from the foot's bend	174.5
three rods' tips above the deck: flush with the dome's outside	43.5
the live rod's tip above the deck, under the dome	42.1
rod holes, three (coaster, deck and roof)	1.8
the live rod's holes, sleeved (coaster and deck)	2.3
each rod's foot under the coaster's top	6.0
motor's bed, height	2.0
the magnet arm above the coaster	32.5
magnet arm, from the spine's side	6.5
magnet from the axis	7.0
reed behind the axis	7.5
reed's top above the coaster	29.5
cell tray's front face behind the axis	9.5
cell tray, across	18.0
cell tray with the cell's bulge, front to back	17.0
cell tray, tall	43.0
cell, diameter	16.6
cell, length	34.1
charger board, length (port at one end)	29.2
charger board, width	19.1
charger board, thick, stripped	4.8
spine, cut length	148.0
spine, outside	4.0
spine's top above the deck	4.0
gallery deck, diameter	38.1
gallery deck, thickness (measure)	3.0
deck's centre hole (as bought)	11.1

bearing wheel's rim (a ring of the 5/8 in tube), outside	15.9
bearing hub's bore (a ring of the 5 mm tube), with play	4.2
bearing wheel, tall	3.0
each spoke, 1 mm rod	14.3
the roof's rim above the deck: the open lantern	34.5
roof (vase cap), diameter	38.1
roof, height (measure)	12.5
shutter drum, outside	15.9
shutter drum wall	0.4
shutter drum, height	35.0
drum's bottom edge above the deck	2.0
the drum's two bars above its bottom edge	0.5
each bar, 1 mm rod	14.6
shutter window, degrees	120.0
shutter window, width on the tube	16.6
shutter window, height	30.0
filament, length	26.0
filament top above the deck	36.5
filament bottom above the deck	10.5
lamp post behind the axis	3.1
lamp post, cut length	42.0
nut's underside above the deck	41.6
ventilator ball	15.9
nipple, length	12.7

MASS

PART	G
brass coaster	80.0
cell and holder, standing at the foot	26.0
motor, on its bed	10.0
magnet arm, magnet, reed, button, wires at the foot	2.1

charger, controller, driver, diode, resistors, wires in the hollow	11.5
four rods, 1.5 mm brass	13.4
spine, 4 mm brass tube	6.9
gallery, turned check ring	26.6
bearing wheel: two tube rings and two 1 mm spokes	0.8
shutter drum and its two bars	5.4
filament, lamp post, two wires	1.5
vase cap	8.2
nut and nipple	7.0
brass ball	17.9
Total, before the ballast	**217**

Centre of mass 81 mm up: it tips at 25°, a 0.3 N push at the ball. With about 80 g of steel washers hot-glued into the coaster's hollow beside the boards: 60 mm, 32°, 0.5 N. The lantern's 67 g ride on four 1.5 mm rods: 0.17 N each against an Euler load of 35 N for a rod fixed at both ends, and the cage sways about 0.08 mm at the deck for every newton of push. The coaster's mass is the listing's; the rest is computed from the drawings or estimated; weigh them.

Pin map, ESP32-C3 SuperMini

Nine pins of the sixteen. The button keeps the family's GPIO10; the cell is read on GPIO4; the boost is switched on GPIO7; no strapping pin is wired. The controller runs from the cell through the diode, as the satellite does.

PIN	JOB	GOES TO
5V	power in: the cell through the SS14, 3.5 to 4.0 V, which the board's own regulator turns into 3.3 V	SS14 cathode; anode to the charger's BAT pad
GND	one ground	charger G · driver GND · reed leg 2 · button leg 2 · the back-right rod's foot (the lamp's return) · the divider's bottom
GPIO4	the cell's voltage, ADC, through two 100 kΩ (half the cell: 2.1 V at full)	the divider's middle
GPIO7	the 5 V boost's enable: high to run the motor and the lamp, low to sleep	charger EN pad
GPIO0	shutter motor, LEDC PWM at 20 kHz (silent), ramped so the boost never sees a step	driver IN1
GPIO1	shutter direction and brake	driver IN2
GPIO3	lamp, LEDC PWM at 5 kHz	driver IN3
GPIO6	index: the reed switch, internal pull-up, closes once a turn	reed leg 1; leg 2 to GND

GPIO10	the button inside the coaster's rim at the front, active low, internal pull-up (the family's pin)	button leg 1; leg 2 to GND
GPIO2, 8, 9	strapping pins: left alone (GPIO8 is the blue LED, held high)	nothing
GPIO5, 20, 21	free	nothing
USB-C	flashing and the console only; the diode keeps its 5 V out of the cell	a computer's cable, in the notch at the back beside the charger's port

CHARGER BOARD PADS

Used: the battery pads where the JST socket was (the tray's leads), `BAT` (to the diode and the divider), `5V` (the boost, to the driver), `EN` (from GPIO7), `G` (ground), the USB-C port (charging). Not used: `VIN`, `VU`, `D+`, `D-`, the terminal block (removed). The charge-rate jumper on the back is cut: 500 mA.

DRIVER BOARD PADS

Used: `VCC` (+) and `GND` (the boost's 5 V), `IN1` `IN2` (channel A, the shutter), `IN3` (channel B, the lamp), `IN4` (tied to GND), `OUT1` `OUT2` (the motor), `OUT3` (the lamp through its resistors and up the back-left rod). `OUT4` is unused: the lamp returns through the back-right rod to ground. The header and the two screw terminals come off; wires go on the pads.

Every wire, and why

NET	FROM	TO	WIRE	WHY
CELL	holder's two leads	the charger's battery pads (where its JST socket was)	the holder's own leads, 60 mm, down through the left hole	The cell, a protected Nitecore 16340, button top, never soldered: it drops into the owned holder standing on the coaster behind the motor, and the holder's leads go down through the coaster's top to the charger's battery pads. The JST socket comes off the charger so the board lies flat in the hollow; the holder is empty while the iron is out. Meter the leads for polarity before the cell goes in.
VBAT	charger BAT pad	SS14 anode · the divider's top	30 AWG, 30 mm	The cell's own voltage, 3.3 to 4.2 V, out of the charger: it feeds the controller through the diode and the divider that measures it. Plugged in, the charger's power path feeds this pad from the cable first and the cell with what is left (the charger page says so), so the lighthouse runs while it charges.
V5C	SS14 cathode (band)	controller 5V pin	30 AWG, 30 mm	The controller's power: the cell less 0.15 V, 3.5 to 4.0 V, which its own regulator turns into 3.3 V, exactly as the satellite is wired. The diode is for the controller's own USB-C: on the SuperMini the 5V pin is VBUS with no diode, so without it a flashing cable would push 5 V into a lithium cell. With it, flash whenever you like.

V5	charger 5V pad (the boost)	driver VCC (+) pad	30 AWG, 40 mm	The 5 V rail for the motor and the lamp, from the charger's TPS61023 boost: 1 A rated, 119 mA asked of it at most in normal running, 70 typical. The boost stalls if a load over 200 mA is switched onto it in one step, so the firmware ramps the motor's PWM up over 0.2 s and never starts the motor and the lamp in the same instant. A jammed shutter would ask 484 mA; the reed's silence catches it first.
EN	GPIO7	charger EN pad	30 AWG, 40 mm	The boost's enable: high while the shutter turns or the lamp is lit, low otherwise, so a sleeping lighthouse costs the cell 20 mA instead of 112. The pad is pulled up to the cell rail on the board (100 kΩ, the satellite's reading); 3.3 V logic drives it either way.
GND	controller GND pin	charger G pad · driver GND pad · reed leg 2 · button leg 2 · the back-right rod's foot · the divider's bottom	30 AWG	One ground. The charger's G, the driver's GND and the controller's GND pins are tied; the lamp returns by way of the back-right rod, soldered to a ground wire at its foot in the hollow. The back-right rod, the coaster, the deck and the roof may all sit at ground; only the back-left rod is live, and it is sleeved.
IN4	driver IN4 pad	driver GND pad	30 AWG, 10 mm	Holds channel B's second input low so OUT3 alone does the switching. A floating input on this chip is undefined. OUT4 is left unconnected: the lamp returns through the frame, not through the half-bridge.
SENSE	charger BAT pad, through 100 kΩ	GPIO4; and GPIO4 through 100 kΩ to GND	two resistors in the hollow, 30 AWG, 30 mm	The cell's voltage, halved: 2.10 V at a full cell, inside the C3's ADC range. 21 µA through the pair, nothing against the cell. Below 3.3 V the firmware dims the lamp, parks the shutter and then shuts the boost off; the cell's own protection is the last word at 2.8 V.
IN1	GPIO0	driver IN1	30 AWG, 30 mm	The shutter's speed: PWM at 20 kHz, above hearing. About 32 % duty gives 8 rpm from a 30 rpm (6 V) gearmotor on 5 V; the firmware trims the duty from the reed's period so a revolution takes 7.5 s whatever the motor's batch. The pin floats for a moment at boot and the spine may twitch once; harmless.
IN2	GPIO1	driver IN2	30 AWG, 30 mm	Direction and brake. Low while IN1 runs: forward, the way that brings the window round to the front after the index. Both high: the bridge shorts the motor and the shutter stops within a degree or two of where the reed said it was. Reverse is there for the parking move and nothing else.
IN3	GPIO3	driver IN3	30 AWG, 30 mm	The lamp's brightness: PWM at 5 kHz into channel B. 70 % by default, 100 % on a long press, 0 when the shutter closes. The driver's channel B is a plain high-side switch here; it never sees a motor.
MOT	driver OUT1, OUT2	motor terminals	2 x 30 AWG, 50 mm, up through the right hole	The gearmotor, about 35 mA turning, 400 mA if the shutter ever jams; the driver is good for 1.5 A a channel and has its own flyback diodes. The motor's two tabs are bent flat outward before soldering so the can stands on its bed; heat-shrink on both.

LAMP	driver OUT3	two 47 Ω in parallel (23.5 Ω), then the back-left rod's foot; at the top a wire from the rod's tip to the filament's + tab	30 AWG, 40 mm in the hollow; the rod itself, 223 mm; 30 AWG, 30 mm under the roof	A 3 V, 100 mA filament from the 5 V rail: 23.5 Ω drops the other two volts and sets 83 mA flat out, inside the rating; each resistor dissipates 81 mW of its 250. PWM takes it to 58 mA for everyday. The back-left rod carries it up: 8.1 mΩ of brass, sleeved in white heat-shrink where it passes the coaster and the deck and booted at its tip, so it touches no other brass. The + tab is the one with the hole in it (the filament page); the diode test lights it faintly. If a sleeve ever failed, 5 V across the resistors would be 532 mW each: warm, not a fire, and the meter check before power catches it.
LAMP-	filament – tab	the lamp post (1 mm brass rod), the nut, a wire to the back-right rod's tip; the rod; a wire from its foot to controller GND	the post, 42 mm; 30 AWG, 30 mm under the roof; the rod; 30 AWG, 40 mm in the hollow	The lamp's return. A rigid filament has a tab at each end, so the lower tab is soldered to the foot of a 1 mm brass rod that stands behind the lamp and is soldered up inside the nipple's bore: that rod holds the lamp and carries its current. A wire from the nut goes to the back-right rod's tip under the roof, and the rod takes it down to the hollow. Solder at both ends of every rod, so no joint depends on glue or a thread for contact.
REED	GPIO6	reed leg 1	30 AWG, 50 mm, up through the right hole	The index. A 3 × 1 mm magnet under a 1 mm brass arm on the spine passes 2 mm over the reed's top end once a revolution and closes it for about 0.29 s. The firmware times revolutions from it, parks the window on it, and reads a jammed shutter from its silence. The glass is 2 mm thick and 14 long: it stands behind the motor on Kapton, its top 29.5 mm above the coaster, in front of the cell.
REED-	reed leg 2	controller GND pin	30 AWG, 50 mm	The reed's return. Its top lead is bent down along the glass in heat-shrink; both wires go down beside the motor to the right hole.
BTN	GPIO10	button leg 1	30 AWG, 40 mm, inside the hollow	Short press: rotation on or off (off parks the window facing you). Long press: lamp off, low, full; off also closes the shutter. Internal pull-up; no resistor.
BTN-	button leg 2	controller GND pin	30 AWG, 40 mm	The button's return. The 6 mm tactile is glued inside the coaster's rim at the front, its plunger through a 3.6 mm hole 3.4 mm up from the bottom edge: press the front of the coaster.
CHG	any USB-C adapter or a computer	charger USB-C, in the notch at the back	the cable	Charging: 500 mA with the board's jumper cut (its default is 1 A, too much for a 950 mAh cell), about 2.3 h from empty, the orange LED while it charges; the chip's own 6 h timer ends it. Plugged in, the power path runs the lighthouse from the cable and fills the cell with the rest. Any phone adapter.
USB	a computer	controller USB-C, in the same notch	the cable	Flashing and the console only, through the second port in the notch; the diode keeps its 5 V out of the cell. The lighthouse runs on its cell while it is flashed.

The budget

MODE	MA FROM THE CELL	NOTE

beacon: shutter turning, lamp at the everyday level	129	70 mA on the 5 V rail through the boost at 88 %, plus the controller; about 7 h on a charge
lamp alone, shutter parked	112	about 8 h
asleep: boost off, controller waiting for the button	20	about 2 days
the worst moment	484	119 mA on the rail with the motor and the lamp both full; the cell is good for 2 A
charging	500	into the cell, the jumper cut; about 2.3 h from empty; the chip's 6 h timer ends it

On the 5 V rail itself: the motor 11 mA average (35 at 32 % duty, about 400 if the shutter jams), the lamp 58 everyday and 83 full (249 mW in the lantern, 81 mW in each resistor), the driver 1. The boost is rated 1 A and stalls only if more than 200 mA is switched onto it in one step, so the firmware ramps. The controller takes its 20 mA from the cell through the diode, not from the rail, so the rail can sleep.

Speed: a 30 rpm (6 V) gearmotor runs about 25 rpm on 5 V; 32 % duty gives 8 rpm, 7.5 s a revolution, a 2.5 s pass of the 120° window. The reed's period closes the loop, so the number holds whatever the motor's batch or the cell's state.

Parts

Rev C sent fourteen lines to buy in three orders. The eight Amazon lines went in on 2026-11-09 (five orders, $122.64 with tax and delivery; that morning a pin vise came off the cart because a cordless drill is owned, and the coaster because the one bought on 2026-11-06 now belongs to this build), and the five Grand Brass lamp parts by the builder's word (none of them is on Amazon; no confirmation mail had come by that evening). The charger is still to buy: that morning's DigiKey order holds the rover's six lines. That evening Rev D dropped two of the Amazon lines: the brass sheet, which the builder cancelled, and the acrylic tube, which had shipped before it was rejected and goes back. The cell, its tray, the diode, the resistors, the controller, the driver, the rods, the tubes and the wire are in a drawer and listed so they are not bought twice. Each part is named by what it is, with the link on the word to the exact product. Prices are what Amazon charged on 2026-11-09.

ORDERED ON AMAZON 2026-11-09: EIGHT LINES IN FIVE ORDERS, $122.64 WITH TAX AND DELIVERY; TWO OF THEM DROPPED THAT EVENING (THE SHEET CANCELLED, THE ACRYLIC TUBE TO RETURN)

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
0 (to return)	clear acrylic tube, 25 mm OD × 21 mm ID × 305 mm	uxcell, Amazon B08ZM5BCQJ	$8.99; ordered 2026-11-09 (order 114-1438198-3243448) and shipped the same day; rejected that evening: return it when it comes 11-11, or keep it as stock	Rev C's lantern glass, dropped in Rev D at the builder's word ("so ugly, never"). The lantern is open: the shutter turns in the air between the rods, and since the glass was a window and not a lens nothing optical is lost.

1 of 2	N20 gearmotor, 6 V, 30 rpm, 3 mm D shaft	Augiimor 2-pack, Amazon B096S475NH	$15.99 the pair; ordered 2026-11-09 (order 114-1438198-3243448), shipped the same day, arriving Wed 11-11	Turns the shutter from the foot of the tower. 12 × 10 × 24 mm body plus a 9 mm shaft, 30 to 40 mA. On 5 V it would run 25 rpm; PWM holds it at 8, one revolution in 7.5 s. It stands shaft-up on a 2 mm bed of hot glue on the coaster's top, its shaft on the axis; the spine slides over the D shaft with a drop of gel glue. Its two tabs bend flat outward so the can sits level. The second motor is the spare. **Incoming check:** 3 mm D shaft, and it runs at 3 V on the bench supply.
1 of 10	LED filament, 26 mm, 3 V 100 mA, warm white 2700 K	UGEFKMQ 10-pack, Amazon B0HBX2YW18, the 26 mm 2700 K variant of the listing whose 38 mm child is the fallback	$25.40 plus $4.98 delivery; ordered 2026-11-09 (order 114-6111517-7569023), arriving Fri 11-27, the last box	The lamp: a strip of LEDs, the kind inside an Edison bulb, with a metal tab at each end; the tab with the hole is + (the 26 mm filament page says so of its own). It stands on the axis, soldered by its lower tab to the lamp post and fed at its upper tab by a wire. 2700 K, not the family's 2200 K spare (amber; a lighthouse lamp is warm white, and the rule is white or warm white light). 83 mA flat out through 23.5 Ω, 249 mW, in the open air. If 26 mm is out of stock, the 38 mm 2700 K pack fits with the shutter cut 12 mm taller and the post 12 longer; nine spare either way. The DLEDMOP listing first named here turned out to be its 18.5 mm item with no size picker, so the cart holds this one; a 26 mm 2700 K pack rated 50 mA is cheaper and sooner but would want one 47 Ω instead of two and give half the light.
1 of 20	glass reed switch, 2 × 14 mm, normally open, 7 to 35 AT	JESSINIE MKA-14103 20-pack, Amazon B0CHF29394	$9.82; ordered 2026-11-09 (order 114-1438198-3243448), shipped the same day, arriving Wed 11-11	The index. Stands behind the motor, in front of the cell, in the open, on Kapton, its top 2 mm under the magnet's path. Pick the most sensitive of the twenty on the bench (the one that closes farthest from the magnet). Glass: never bend a lead at the seal.
1 of 100	neodymium disc magnets, 3 × 1 mm, N52	Amazon B0BQ3LPGZ1 (100 or more a pack)	$19.49; ordered 2026-11-09 (order 114-1438198-3243448), shipped the same day, arriving Wed 11-11	The index magnet, glued under the tip of the 1 mm brass arm on the spine, 7 mm from the axis, opposite the window's middle so the reed fires as the window faces front. One is enough; a second stacked on it doubles the field if the reed is lazy. It shows as a black dot under a thin brass arm.

0 (cancelled)	brass sheet, 0.015 in (0.4 mm), 4 × 10 in	K&S 252, Amazon B0006N6WJE	$5.31 plus $9.49 delivery; ordered 2026-11-09 (order 114-5648832-0318640); cancelled by the builder that evening (confirm with Amazon's cancellation mail)	Rev C's keeper's house and its three sheet-cut discs. Dropped: the builder does not fold brass sheet round components, so Rev D has no house, and the shutter's floor, the magnet's collar and the deck's bushing are made from the 1 mm rod and the tubes already on the bench (below).
1	brass round tube, 5/8 in OD × 0.014 in wall × 12 in	K&S 8143, Amazon B07DDFNNXY	$9.89; ordered 2026-11-09 (order 114-7188491-2312247), arriving 11-18 to 11-20	The shutter: a 35 mm length with a 120° window sawn out (16.6 mm of the 49.9 mm circumference, 30 mm tall); two 1 mm bars across its bottom, 0.5 mm up, soldered to the spine's flanks and to the wall, are its floor. A second piece, a 3 mm ring from the same tube, is the rim of the deck's bearing wheel. 0.36 mm wall: saw gently, file square, polish. The template sheet has the window at actual size.
0 (dropped)	pin vise with 0.5 to 3.0 mm twist bits	Amazon B09WRQPN58; dropped from the cart 2026-11-09 at the builder's word	none	Dropped: a 20 V cordless drill is owned (bought 2026-10-11, missing from the inventory until 2026-11-09), so the builder took this set out of the cart on 2026-11-09. The holes go to that drill at its slowest, with metric bits from its case or a small bit set if the case has none under 3 mm: the coaster's four rod holes (1.8, one 2.3), two wire holes (3.5) and the button's 3.6 in the rim, the deck's four rod holes, the roof's three 1.8 holes. Centre-punch first; brass grabs a twist bit, so a slow speed and a light hand.
1 tube	gel super glue, 4 g	Loctite Ultra Gel Control, Amazon B003Y49R7G (it was waiting in the builder's Saved for later; the hologram's and the rover's tube)	$3.29; ordered 2026-11-09 (order 114-1965907-0721062), arriving Sat 11-14	One tube for three builds. Here: the rods' feet under the coaster, the deck to the rods, the wheel's rim to the deck, the magnet under its arm, the reed to the coaster, the spine on the D shaft, the tray to the coaster, the button inside the rim. No epoxy is owned and none is needed.

GRAND BRASS LAMP PARTS: FIVE LINES, ONE ORDER, PLACED BY THE BUILDER ON 2026-11-09 BY THEIR WORD (NO CONFIRMATION MAIL HAD COME BY THAT EVENING, SO NO ORDER NUMBER HERE; NONE OF THESE IS ON AMAZON)

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1	turned brass check ring, 1-1/2 in, 1/8 IP slip (7/16 in hole)	Grand Brass CRT1-12X8, unfinished	about $1.60	The gallery deck: a solid turned brass disc, Ø38.1, about 3 mm thick (measure it; the drawing assumes 3), with an 11 mm hole already in its middle that the spine passes through. Threaded down the four rods to 168 mm above the coaster and glued to them. Four holes to drill at r 14.5: 1.8 mm for three rods, 2.3 for the sleeved live one. No railing: the slab is the gallery.
1	spun brass vase cap, 1-1/2 in, 1/8 IP slip hole	Grand Brass VA01-12, unfinished (the vase-cap listing, 1-1/2 in)	about $2.50	The roof. A shallow spun dome Ø38.1, about 12.5 mm tall (measure; drawn 12.5), the same diameter as the deck, so the lantern is an open drum of air between two brass discs. Three Ø1.8 holes drilled in its dome at r 14.5: the three grounded rods pass through and are soldered flush from outside, so the roof is part of the frame; the live rod ends 42 mm above the deck under it, booted. Its centre hole takes the nipple.
1	brass ball, 5/8 in, 1/8 IP blind tap	Grand Brass BB0-58X18, unfinished brass (listed beside the nickel one)	about $1	The ventilator ball on the roof, the one silhouette a lighthouse needs after its lantern. Solid, 18 g, screwed onto the nipple until it meets the cap. Unfinished brass, polished by hand.
1	hollow brass nipple, 1/2 in long, 1/8 IP	Grand Brass NI0-12X18BR	about $1	Through the cap's hole: the ball on top, the nut underneath. 12.7 mm long; file it shorter if the ball will not seat. The lamp post is soldered up inside its bore, 5 mm deep.
1	beaded brass nut, 1/8 IP, 9/16 in × 3/16 in	Grand Brass, unfinished	about $1	Inside the cap on the nipple: it clamps the roof's hardware, and the lamp's return wire is soldered to it. The nut, the nipple and the post are one piece of brass to the meter.

STILL TO BUY: THE CHARGER, ONE LINE (ADAFRUIT, OR DIGIKEY 1528-6106-ND; THE DIGIKEY ORDER OF 2026-11-09 WAS THE ROVER'S AND DOES NOT HOLD IT)

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	bq25185 USB / DC / solar charger with 5 V boost	Adafruit 6106 or DigiKey 1528-6106-ND, in stock at both on 2026-11-09	$8.95; not ordered as of 2026-11-09 evening	The cell's charger and the 5 V rail in one 29 × 19 mm board, the family's charger (Cyber Dawn's and the satellite's; both of theirs are spoken for, so this is a third). USB-C in; power path, so plugged in the lighthouse runs from the cable and the cell fills with the rest; the charge-rate jumper on the back cut for 500 mA (its default is 1 A); a TPS61023 boost, 5 V at 1 A, that the firmware switches off through EN when nothing moves. The screw terminal and the JST socket come off so it lies 4.8 mm flat in the hollow with its port in the notch (the pinouts name every pad). **Incoming check:** cut the jumper before the cell meets it.

OWNED: ELECTRONICS AND POWER

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1 of 13	Nitecore 16340 cell, 950 mAh, protected, button top	Nitecore NL169, IMR Batteries orders of 2026-10-16	owned	The battery, the family's cell: 16.6 × 34.1 mm, 20 g, protection under the wrap, 2 A limit, 4.20 V termination. Standing in the holder behind the motor, in the open; it lifts straight out. About 7 h of beacon with the lamp at 70 %, 8 h of lamp alone, 2 days asleep; 2.3 h to charge. **Incoming check:** 3.6 to 4.1 V on the meter, button top +; nothing else. Never solder to it.
1 of 10	16340 holder with leads	ACEIRMC, Amazon B0CRGK889F (Cyber Dawn's stock)	owned	A 43 × 18 × 14 mm black plastic tray with two spring contacts and two leads, standing on end on the coaster's top behind the motor on a dot of glue, its leads down through the left hole to the charger's battery pads. The cell drops in from the top. Black is a colour the rule allows in the open. Meter the leads for polarity before the cell goes in.
1 of 13	ESP32-C3 SuperMini controller	Amazon B0F888JQ91 (Pocket AI stock; 10 spare)	owned	The brain, in the coaster's hollow, glued up to the ceiling on Kapton with its USB-C in the notch at the back beside the charger's: flashing and the console. It runs from the cell through the diode. No Wi-Fi in this build; the button in the rim is the whole interface. Gate it with `esptool flash_id` as before; nothing is removed from it.
1 of 5	MX1508 dual H-bridge driver	Aideepen 5-pack, Amazon B075S368Y2 (the camera Pocket AI's stock; 4 spare)	owned	One board for both moving things, on the boost's 5 V: channel A turns the shutter (speed, brake, the parking move), channel B switches the lamp through its resistors. 2 to 10 V in, 1.5 A a channel, inputs that read 3.3 V as high, flyback diodes inside. The screw terminals and the header come off and wires go on the pads, so it lies 3 mm flat in the hollow at the front.
1 of 100	SS14 Schottky diode	Chanzon, Amazon B079KJ46ZS	owned	Between the charger's BAT pad and the controller's 5V pin, band toward the controller: 0.15 V lost, and a flashing cable's 5 V can never reach the cell. The satellite's wiring, unchanged.
2 + 2	47 Ω and 100 kΩ resistors, 1/4 W	the resistor kit	owned	Two 47 Ω in parallel, 23.5 Ω, in series with the lamp, in heat-shrink in the hollow (81 mW each at full). Two 100 kΩ as the cell-voltage divider into GPIO4. The only resistors in the build.
1	tactile button, 6 × 6 mm	the button kit	owned	The one control, glued inside the coaster's rim at the front with its plunger through a 3.6 mm hole 3.4 mm up from the bottom edge: press the front of the coaster. Nothing shows but the plunger's tip.
about 1 m	30 AWG silicone wire	CBAZY kit	owned	Two to the motor and two to the reed up through the right hole, the holder's own two leads down the left, two to the button inside the rim, eight between the three boards and the diode and the divider, two to the rods' feet, two under the roof. White where a wire could be seen.
	Kapton tape, white heat-shrink, hot glue, 63/37 solder and flux	bench stock	owned	Kapton under the boards, the reed and the live rod's foot; white heat-shrink on the live rod where it passes the coaster and the deck, as a boot on its tip, and on every joint near brass; hot glue for the motor's bed, the washers, the button and the boards; solder for the shutter's bars, the magnet arm, the wheel, the lamp post, the rods' tips in the roof and their feet.

1	USB-C adapter, 1 A or more, and a cable	owned	owned	Any phone adapter, for the charger's port at the back. The lighthouse takes 500 mA to charge and runs from the cable while it does.

OWNED: BRASS

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	brass coaster, 75 × 8 mm, flat top	Amazon 114-8443990-0476239, ordered 2026-11-06 ($12.99), delivered 2026-11-09 to the mail room (Amazon's mail), in one package with the drone's heat tape	owned	The base and the electronics box in one. Bought on 2026-11-06 as the satellite's base and given to this build on 2026-11-09 at the builder's word (the satellite buys its own again when it is built); the same part under both, so the two desk pieces match. 80 g, which says a top about 1.3 mm thick over a hollow about 6.7 deep: the four rods pass through the top and bend under it; the charger, the controller and the driver glue up to the hollow's ceiling; the two USB-C ports share a 40 × 6.5 notch sawn in the rim at the back; about 80 g of steel washers fill the rest. Alone it tips at 25° (0.3 N at the ball); weighted, 32° (0.5 N). **Incoming check:** the hollow's depth (the controller wants 5.3 mm, the charger 4.8, the button's body 6; if it is shallower the boards do not fit and the base must change); a magnet must not stick to the top.
4 rods	brass rod, 1.5 mm × 300 mm	K&S 9862, 5 a pack; about 90 rods owned	owned	The cage: four rods cut 223 mm, each from the coaster through the deck to the roof, bent to a 6 mm foot under the coaster's top and nudged 4.3° to vertical just under the deck; three pass through the roof's dome and are soldered flush, the live one is trimmed to end 42 mm above the deck under it. Two of them carry the lamp's current. Solid rod, not the 1.5 mm tube: about 30 % stiffer, and a conductor you can solder to.
about 150 mm	brass tube, 4 mm OD × 0.5 mm wall (from the 2 to 7 mm set, 200 mm)	M METERXITY set, 3 owned	owned	The spine: one piece 148 mm long, a slip fit over the motor's 3 mm D shaft with a drop of gel glue, up through the hub of the deck's wheel, soldered to the shutter's two bars and ending 0 mm above them. It turns, so it is the one moving line in the tower. From the same set, a 3 mm ring of the 5 mm tube is the wheel's hub: the 4 mm spine telescopes into it, which is what the set is made for.
about 106 mm	brass rod, 1.0 mm × 300 mm	K&S 9861, 10 owned	owned	The lamp post: 42 mm, soldered 5 mm up inside the nipple's bore, down behind the lamp at r 3.1, bent forward at the bottom and soldered to the filament's lower tab; it holds the lamp and is its return. And the spine's hardware, 64 mm in all: two bars of 14.6 across the shutter's bottom, two spokes of 14.3 in the deck's wheel, one arm of 6.5 for the magnet. Nothing in Rev D is painted.

What the firmware needs

Not written yet; listed so the pin map above is known to be enough.

- ESP-IDF 6 for the esp32c3, a new small project beside the hologram's. No Wi-Fi: the console on USB Serial/JTAG, the button the whole interface.

- GPIO7 high to wake the boost before anything moves, low again when the shutter is parked and the lamp is off; the boost stalls over 200 mA at switch-on, so the motor's PWM ramps from 0 to its duty over 0.2 s and the lamp fades up after it.
- LEDC: GPIO0 at 20 kHz for the motor (silent), GPIO3 at 5 kHz for the lamp. GPIO1 low for forward, high with GPIO0 high to brake. All three driven low first thing at boot.
- The reed on GPIO6, interrupt on the falling edge, 5 ms debounce: period of the last revolution; a PI loop trims the motor duty to hold 7.5 s; no pulse for twice that with the motor on = jammed, motor off, boost off.
- Parking: on rotation off, run until the reed closes, then brake: the window faces front (the magnet sits opposite the window, over the reed behind the motor). On lamp off, run from the index for half a period, then brake: the shutter is closed. On lamp on, open again. A small offset in NVS trims the park angle after the first bench run.
- The button on GPIO10: short press rotation on/off; long press (1 s) lamp off, low, full. The lamp blinks once to say so.
- The cell on GPIO4 through the divider, read once a minute: below 3.5 V the lamp drops to low and blinks twice; below 3.3 V the shutter parks, the lamp goes out, the boost goes off, and the controller sleeps until the button or a higher reading.
- First-power measurements the record wants: the current from the cell in each mode (expected about 129 mA beacon, 112 lamp, 20 asleep), the lamp's current at full (about 83 mA), the revolution period from the reed, and the charge current with the jumper cut (about 500 mA).

Before power, with the meter

- [ ] Before the cell meets anything: the holder's leads for polarity on the meter with the cell in (3.6 to 4.1 V, button top +); the charger's jumper cut; the filament's polarity with the diode test (it lights faintly; the tab with the hole is +).
- [ ] With the boards wired on the bench, the cell in and the lantern not yet up: the cell's voltage on the charger's BAT pad, 3.5 to 4.0 V on the controller's 5V pin (the diode's drop), 5.0 V on the driver's VCC with GPIO7 high and nothing with it low, half the cell's voltage on GPIO4; then the lamp channel at full gives 2.9 to 3.1 V across the filament, about 83 mA, the motor channel turns the spine, the brake stops it, and the reed closes once a turn. Plug the charger's cable: the orange LED, and the lighthouse still runs.
- [ ] The live rod alone: back-left rod to the coaster open, to the deck open, to the roof open, before its wires go on. Then back-right rod's foot to the nut: a short, through the solder at both ends. Nothing else bare against brass: the motor's tabs, the reed's leads, the holder's leads and both wires under the roof in heat-shrink. Everything else is a supplier spec and is not metered.

Rules this build follows

- Design first: sleek, few lines (four rods and a spine; no rings, no braces, no glass, no house); wires in the hollow, two of them the rods themselves; no closed surface at all.
- Colour: brass and warm white; black only in the pad under the coaster, the cell's wrap and its tray, in the open at the foot as the rule allows; the boards in the hollow; the filament's yellow phosphor hidden by the shutter when off; white heat-shrink where a sleeve shows.
- No clear acrylic and no folded brass sheet, at the builder's word of 2026-11-09 evening: the lantern is open and the three small brass pieces are rod and tube.
- Power drawn on both sheets: the cell stands where the form sheet shows it, and the wiring sheet reads cell, charger, controller, driver, loads.
- Component names with the link on the word; no product codes in prose.

- Owned parts stay off the shopping list: the cell, the tray, the diode, the resistors, the controller and the driver are family stock, the rods and the tubes are shared stock.
- Every wire on the sheet is traceable pin to pin; the generator checks clearance and counts crossings (zero).
- Brass at the builder's word for this build (2026-11-07); the family's rule since the drone is stainless.
- Meter only what matters: three checks before power, one of them the live rod against every other piece of brass, one the tray's polarity before the cell meets anything.

Status

Rev C was approved by the builder on 2026-11-09 as drawn. That evening the builder cancelled the brass sheet ("very hard to wrap round electric components"), rejected the acrylic tube ("so ugly, never") and any plastic house ("just use the wire frame as a lighthouse, nothing else"), so Rev D, drawn the same evening, removes the glass and the house and makes the three small brass pieces from rod and tube; everything else stands. **Rev D awaits the builder's approval.** The same three sizes are assumed and must be measured when the parts come: the check ring's thickness (3 mm), the vase cap's height (12.5 mm) and the coaster's hollow (about 6.7 deep under a 1.3 mm top, from its 80 g; the coaster arrived 11-09); the rods' tips and whether the boards fit the hollow follow from them, and the script recomputes them. Step 2, the science and engineering behind the sheet, is published as Desk Lighthouse Physics and written to `docs/PHYSICS.md`, revised for the open lantern. Orders: the eight Amazon lines of 2026-11-09 less the sheet (cancelled) and the acrylic tube (to return) are this build's; the five Grand Brass lamp parts by the builder's word; the charger still to buy. Next: the yes or the changes, the charger, the three measurements, then the firmware.

Built by `docs/pages/gen/build_page.py`, which also writes `docs/FINAL_BUILD.md` and the PDFs; the dimensions, pins, nets, budget and parts live in that script. The wiring sheet's clearance check passes, and the clearances the drawing relies on (shutter inside the rods, post in the shutter, three rods through the roof and the live one under it, the magnet arm clear of the tray, the wheel over the deck's hole, the boards in the hollow and their ports in the notch, the button inside the rim, the boost under its switch-on limit) are asserted. Sources read 2026-11-07 to 2026-11-09: the reference page, the Amazon, Grand Brass and Adafruit listings, the charger's pinouts page, the filament page, the SuperMini notes, the MX1508 and SSD1306 datasheets, the family's inventories and the satellite's record. Nothing measured.